

ANALYSTLAB AFRICA
EXPERIENCE LAB INTERNSHIP PROGRAMME
HEALTHCONNECT CLINIC EXPERIENCE LAB
WEEK 7 UPDATED RISK & DEPENDENCY REGISTER

Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Field	Detail
Project:	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Programme:	AnalystLab Africa Experience Lab
Workstream:	Project Management
Phase:	Week 7 — Testing, Refinement & End-to-End Validation
Version:	3.0
Status:	Controlled Working Register
Register Owner:	Project Management
Review Basis:	Week 7 testing evidence, integration findings, validated outputs, outstanding dependencies and Week 8 readiness conditions

1. WEEK 7 RISK CONTROL POSITION

Week 7 represents a shift from prospective risk identification to evidence-based exposure reassessment.

The Week 6 register established the principal delivery risks and dependencies. Testing, refinement and cross-track integration during Week 7 have now provided evidence to determine whether those exposures have been reduced, confirmed, elevated or reframed.

This register therefore records the current project exposure, rather than simply carrying forward the previous reporting position.

Its control boundary is deliberate:

- Testing Activity Tracker — controls test execution and retest evidence.
- Integration & Issue Log — controls specific interface conditions, blockers and corrective actions.
- Risk & Dependency Register — controls project-level exposure, treatment, residual risk and escalation.
- Decision Log — controls formal acceptance and governance decisions.
- Updated Timeline — controls schedule and sequencing consequences.

Week 7 control principle: Risk exposure is determined by current evidence and downstream consequence, not by historical status alone.

2. RISK ASSESSMENT METHOD

Risk statement structure

Each risk is expressed as:



This distinguishes the underlying condition from the event that may occur and the consequence to delivery or acceptance.

Exposure convention

Dimension	Rating	Interpretation
Likelihood	Low / Medium / High	Probability of the risk event occurring or remaining material within the current phase
Impact	Low / Medium / High / Critical	Consequence to scope, quality, integration, safety, acceptance, schedule or readiness
Exposure	Low / Medium / High / Critical	Overall project significance after considering likelihood, impact and critical-path relevance

Critical-path relevance can elevate an exposure where an otherwise contained condition prevents a required integration or readiness gate.

3. UPDATED RISK REGISTER

Each risk record below captures the cause→event→impact statement, Week 7 exposure change, assessed likelihood/impact/exposure, treatment, escalation trigger, owner, residual exposure and status.

R-001 — DS → ML Integration	
Risk Statement (Cause → Event → Impact)	If the final DS-selected candidate and its input specification are not formally transferred and reconciled, ML may validate a technically compatible candidate that is not the actual approved DS output, creating a model-integration acceptance gap.
Week 7 Exposure Change	Elevated — candidate defined; actual final transfer and receipt not evidenced.
Likelihood / Impact / Exposure	High / Critical / Critical
Treatment / Control	Reduce: controlled candidate handoff; confirm model identity/version; reconcile 33-feature schema, ordering and preprocessing; execute the actual candidate through ML.

R-001 — DS → ML Integration	
Trigger / Escalation Condition	Candidate cannot be executed or receiving track cannot confirm candidate identity/configuration before integration gate.
Owner	DS / ML
Residual Exposure	High
Status	Open

R-002 — Operationalisation	
Risk Statement (Cause → Event → Impact)	If model scores are not converted into an accepted threshold, decision rule and intervention pathway, predictive performance may not translate into an actionable HealthConnect workflow, leaving operational value unvalidated.
Week 7 Exposure Change	Elevated — threshold evidence exists; operational acceptance and score-to-action validation remain outstanding.
Likelihood / Impact / Exposure	High / Critical / Critical
Treatment / Control	Reduce: document threshold decision; define risk classification; map classification to intervention; validate the complete score→decision→intervention sequence.
Trigger / Escalation Condition	No accepted threshold, undefined action for a risk class, or inability to complete the operational workflow.
Owner	PM / DS / ML
Residual Exposure	High
Status	Open

R-003 — Analytics Evidence	
Risk Statement (Cause → Event → Impact)	If dashboard metrics and analytical findings are not reconciled and formally dispositioned, downstream decisions may rely on evidence whose calculations, definitions or interpretation have not been fully accepted.
Week 7 Exposure Change	Confirmed / Reframed — dashboard and EDA available; formal reconciliation and downstream disposition incomplete.
Likelihood / Impact / Exposure	Medium / High / High
Treatment / Control	Reduce: KPI reconciliation; runtime filter/interaction validation; confirm metric definitions; record downstream disposition of material findings.
Trigger / Escalation Condition	Material discrepancy, failed interaction, or material finding without documented disposition.
Owner	Analytics / PM

R-003 — Analytics Evidence	
Residual Exposure	Medium
Status	Open

R-004 — Model Generalisation	
Risk Statement (Cause → Event → Impact)	If observed performance limitations are not incorporated into model acceptance and monitoring controls, current validation evidence may be interpreted beyond its demonstrated generalisability.
Week 7 Exposure Change	Confirmed — 5-fold ROC-AUC 0.6771 ± 0.0090 ; train/test ROC-AUC gap 0.0571; segment-level limitations identified.
Likelihood / Impact / Exposure	Medium / High / High
Treatment / Control	Reduce: document limitations; retain threshold sensitivity evidence; assess segment behaviour; define appropriate monitoring and future validation requirements.
Trigger / Escalation Condition	Performance limitation materially affects intended use or acceptance assumptions are being treated as established evidence.
Owner	DS / PM
Residual Exposure	Medium
Status	Open

R-005 — ML Technical → Operational Acceptance	
Risk Statement (Cause → Event → Impact)	If successful technical ML testing is treated as evidence of complete solution readiness, downstream operational conditions may remain unvalidated while the component is incorrectly considered integrated.
Week 7 Exposure Change	Reframed / Partially Reduced — technical evidence substantially strengthened: 69/69 automated tests passed; 948/948 records scored; reproducibility, negative scenarios and backtesting completed.
Likelihood / Impact / Exposure	Medium / High / High
Treatment / Control	Reduce: maintain separate technical and operational acceptance criteria; require downstream validation before readiness status changes.
Trigger / Escalation Condition	Component marked ready while threshold, decision logic or intervention pathway remains unresolved.
Owner	ML / PM
Residual Exposure	Medium

R-005 — ML Technical → Operational Acceptance	
Status	Open

R-006 — GenAI Validation Reliability	
Risk Statement (Cause → Event → Impact)	If external service instability prevents planned scenario execution, required GenAI behaviour may remain insufficiently evidenced at the point of acceptance.
Week 7 Exposure Change	Elevated — two planned scenarios blocked by Gemini 503/high-demand conditions.
Likelihood / Impact / Exposure	High / High / High
Treatment / Control	Reduce: re-execute blocked scenarios; preserve blocked status as untested; maintain a controlled retest window; distinguish service unavailability from functional failure.
Trigger / Escalation Condition	Required validation remains blocked near a readiness gate or repeated service instability prevents adequate evidence collection.
Owner	GenAI / PM
Residual Exposure	Medium
Status	Blocked / Open

R-007 — GenAI Safety & Governance	
Risk Statement (Cause → Event → Impact)	If emergency, privacy and unsupported-information boundaries are not sufficiently exercised across representative scenarios, the GenAI component may enter integrated validation with incomplete safety evidence.
Week 7 Exposure Change	Elevated — selected boundaries tested; broader indirect emergency and privacy-sensitive scenarios remain outstanding.
Likelihood / Impact / Exposure	Medium / Critical / Critical
Treatment / Control	Reduce: expand boundary scenarios; validate KB grounding; test indirect emergency language and privacy-sensitive interactions; maintain human-in-the-loop escalation.
Trigger / Escalation Condition	Safety/privacy scenario remains untested or fails at an acceptance gate; unsupported information or escalation failure is observed.
Owner	GenAI / PM
Residual Exposure	High
Status	Open

R-008 — Cross-Track Acceptance	
Risk Statement (Cause → Event → Impact)	If workstream outputs cross interfaces without explicit receipt, validation and disposition, individual completion may be mistaken for integrated acceptance, creating hidden downstream dependency exposure.
Week 7 Exposure Change	Elevated — Analytics→DS disposition and DS→ML acceptance remain incomplete.
Likelihood / Impact / Exposure	High / High / High
Treatment / Control	Reduce: controlled handoff; receiving-track acceptance condition; evidence of receipt, resulting use/change and verification; unresolved critical interfaces escalated.
Trigger / Escalation Condition	Upstream output marked complete while receiving track cannot confirm usability or acceptance.
Owner	PM / All Tracks
Residual Exposure	Medium
Status	Open

R-009 — End-to-End Readiness	
Risk Statement (Cause → Event → Impact)	If critical component interfaces are not validated as one HealthConnect workflow, successful individual workstream outputs may fail to demonstrate a functioning integrated solution by the Week 8 gate.
Week 7 Exposure Change	Confirmed / Critical — component validation is substantial; full E2E workflow remains unvalidated.
Likelihood / Impact / Exposure	High / Critical / Critical
Treatment / Control	Reduce: clear critical interfaces; execute E2E validation; reconcile component interactions; document exceptions and residual limitations before readiness decision.
Trigger / Escalation Condition	E2E scenario cannot complete, critical interface remains unresolved, or material limitation lacks disposition before Week 8 gate.
Owner	PM / All Tracks
Residual Exposure	High
Status	Open

4. WEEK 7 EXPOSURE MOVEMENT

This control records what Week 7 evidence changed, avoiding simple duplication of the full risk narratives.

Risk	Previous Position	Week 7 Evidence	Current Treatment
R-001 DS→ML	Integration dependency	Candidate and 33-feature structure established; actual candidate transfer not evidenced	Elevated
R-002 Operationalisation	Future acceptance condition	Threshold testing progressed; operational decision/intervention pathway remains unvalidated	Elevated
R-003 Analytics Evidence	Analytical foundation	Dashboard available; reconciliation and downstream disposition incomplete	Confirmed / Reframed
R-004 Generalisation	Model-validation consideration	Quantitative validation and segment limitations now available	Confirmed
R-005 ML Acceptance	Technical integration concern	Strong technical validation separates component readiness from operational readiness	Reframed / Partially Reduced
R-006 GenAI Reliability	External dependency	Two planned tests blocked by service availability	Elevated
R-007 GenAI Safety	Validation requirement	Selected boundaries tested; privacy/broader emergency coverage incomplete	Elevated
R-008 Cross-Track Acceptance	Coordination dependency	Actual interface acceptance gaps remain visible	Elevated
R-009 E2E Readiness	Future integration requirement	No full integrated workflow validation yet	Critical / Confirmed

5. ACTIVE DEPENDENCY REGISTER

A dependency is recorded where one deliverable, decision, condition or service is required before another activity can proceed or be accepted.

ID	Dependency	Required Upstream Condition	Downstream Effect	State	Owner	Control Point
DEP-01	DS candidate transfer	Final candidate identity/version confirmed	Actual ML execution	Outstanding	DS / ML	Before candidate integration
DEP-02	Feature/schema compatibility	33-feature structure and preprocessing reconciled	ML input compatibility	Outstanding	DS / ML	Before candidate execution
DEP-03	Threshold acceptance	Model performance evidence assessed	Operational classification	Outstanding	DS / PM	Before E2E

ID	Dependency	Required Upstream Condition	Downstream Effect	State	Owner	Control Point
DEP-04	Decision/intervention mapping	Classification rule accepted	Operational response	Outstanding	PM / ML	Before E2E
DEP-05	Dashboard acceptance	KPI and interaction validation completed	Accepted analytical evidence	Partially evidenced	Analytics / PM	Week 7 closure
DEP-06	Analytics→DS disposition	Findings received and assessed	Traceable model/decision input	Partially evidenced	Analytics / DS	Before model acceptance
DEP-07	GenAI service availability	External service accessible	Remaining scenario execution	Blocked for two scenarios	GenAI	Before final scenario closure
DEP-08	Critical interface clearance	Upstream acceptance conditions satisfied	E2E validation	Outstanding	PM / All Tracks	Before Week 8 integration

6. CRITICAL READINESS DEPENDENCY PATH

The current critical dependency path is:



This sequence is a control dependency, not a statement that every activity must be performed sequentially. Work may proceed in parallel where technically possible; however, downstream acceptance cannot be confirmed until its prerequisite condition has been evidenced.

7. ESCALATION & EXCEPTION CONTROL

Escalation is triggered by consequence and control-point proximity, not simply by an item being open.

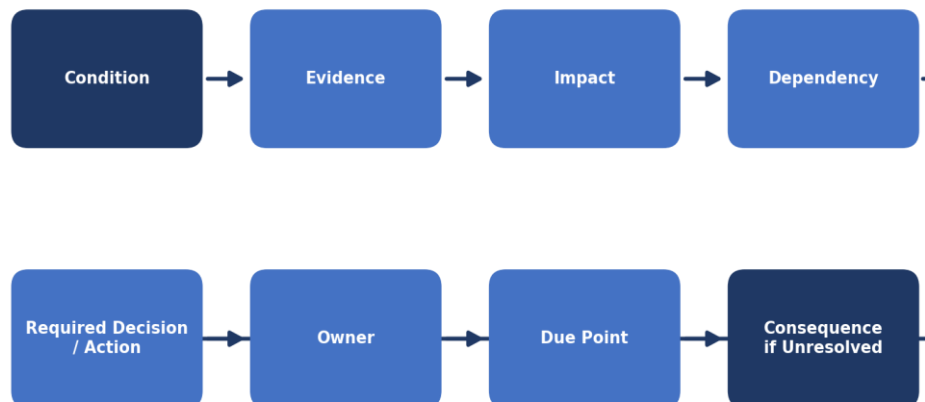
Mandatory escalation conditions

Trigger	Threshold for Escalation	Required PM Control
Critical-path blockage	Dependency prevents DS→ML, operationalisation or E2E progress	Immediate impact assessment, owner confirmation and recovery action
Gate threat	Resolution cannot reasonably be evidenced before the applicable acceptance gate	Escalate for decision, recovery or controlled re-planning

Trigger	Threshold for Escalation	Required PM Control
Safety / privacy exposure	Required GenAI safety/privacy validation is untested, failed or materially inconclusive	Withhold affected acceptance condition pending disposition
Interface rejection	Receiving track cannot reproduce, use or validate upstream output	Initiate cross-track resolution and maintain dependency visibility
Acceptance conflict	Tracks disagree on candidate, schema, metric, threshold or expected behaviour	Formal decision review and documented disposition
Evidence insufficiency	Completion/acceptance is claimed without required objective evidence	Suspend acceptance and identify evidence requirement
External constraint	Service availability materially prevents required testing or compresses the validation window	Escalate constraint, recovery window and residual schedule exposure
Material change	Proposed change affects validated assumptions, interfaces, model inputs or acceptance criteria	Impact assessment before implementation
Residual exposure above tolerance	Mitigation does not reduce exposure sufficiently for the relevant gate	Additional treatment or explicit risk acceptance decision

Escalation information standard

Every escalated item should present:



Escalation path



Escalation does not transfer accountability automatically. The originating owner remains accountable for corrective action unless responsibility is formally reassigned.

8. RISK RESPONSE & CLOSURE CONTROL

Risk treatment shall follow the principle of evidence-backed reduction rather than activity completion.

A risk may move toward closure only when:

1. The treatment has been implemented;
2. The affected condition has been re-tested or objectively verified where applicable;
3. The expected downstream state has been demonstrated;
4. Any remaining limitation is documented;
5. Downstream consequences have been assessed; and
6. Residual exposure is acceptable for the relevant project gate.

Where exposure remains material but is consciously accepted, the register shall record:

Residual Risk + Acceptance Basis + Decision Owner + Validity/Review Point

An action marked complete therefore does not automatically mean the associated risk is closed.

9. GOVERNANCE CROSS-REFERENCE

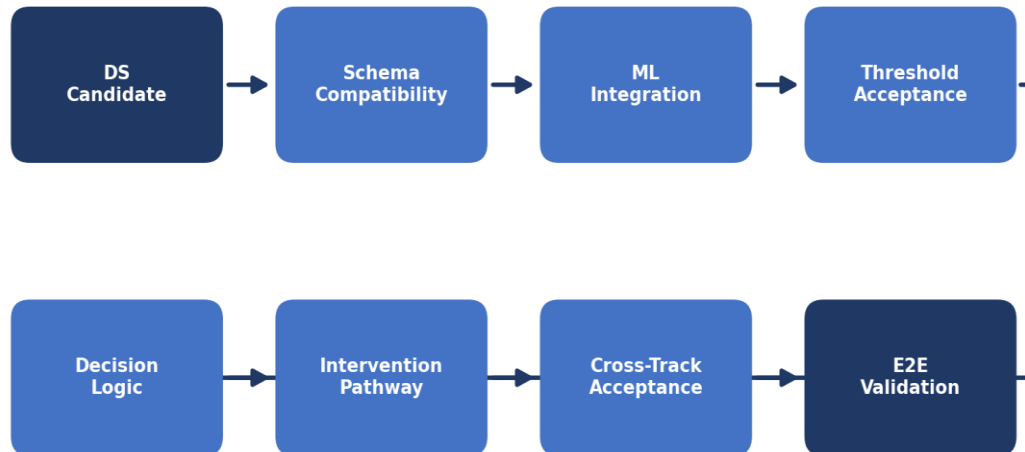
Control Artifact	Primary Purpose	Relationship to This Register
Testing Activity Tracker	Execution, results, findings and retesting	Supplies evidence for risk reassessment
Integration & Issue Log	Interface conditions, blockers and corrective action	Supplies material conditions requiring project-level risk treatment
Risk & Dependency Register	Exposure, treatment, dependency significance and residual risk	Primary project-level risk control
Decision Log	Formal decisions and acceptance	Records decisions arising from risk exposure
Updated Timeline	Sequencing and schedule impact	Reflects material risk-driven schedule changes
Week 8 Readiness Assessment	Gate readiness against acceptance conditions	Uses unresolved residual exposure and dependency status

10. WEEK 7 CONTROL CONCLUSION

Week 7 has produced materially stronger evidence around individual component performance and testing, particularly within ML technical validation and selected GenAI boundaries. At the same time, the testing and integration process

has exposed dependencies that cannot be resolved by treating workstream completion as equivalent to solution acceptance.

The principal project exposure has therefore shifted toward interface acceptance and operational integration:



The project should therefore carry forward conditional readiness, with Week 8 integration dependent on objective evidence that the critical interfaces have been accepted and the remaining residual exposures are either reduced to an acceptable level or formally dispositioned.

No risk is considered closed solely because its originating activity has been completed; closure remains dependent on evidence, downstream effect and residual exposure.